What Stats Actually Predict Winning in the NBA?
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Project Topic: This project analyzes NBA player and team data to determine which statistics are most strongly associated with winning.

Project Objectives

- Identify whether offensive or defensive efficiency is a stronger predictor of winning.
- Measure how strongly three-point shooting relates to team success.
- Test whether teams with high-scoring stars are also the most successful teams.
- Combine player stats, team metrics, and shooting profile data into one analysis.

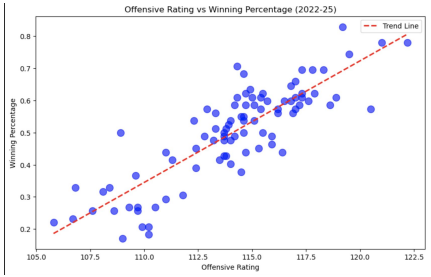
Data / Methods

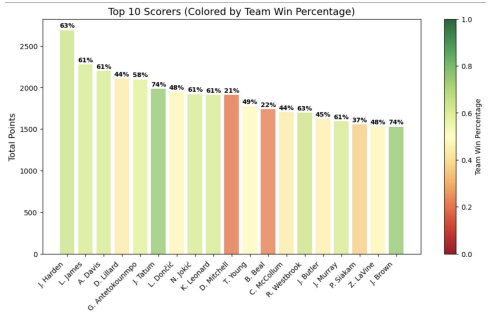
- Downloaded player dataset from sports-statistics.com
- Team data collected with nba_api from nba.com
- Team shooting profile data collected from BasketballReference.com
- Cleaned team names, handled missing values, and merged datasets by team and season



Key Insights

- Offensive efficiency was a stronger predictor of winning than defensive metrics with a +0.028 vs -0.027 linear regression coefficient (meaning better offence helps more than worse defense hurts)
- Three-point shooting does matter, but accuracy is what truly drives success, volume alone has almost no impact.
- Having a high-volume scorer helps, but it doesn't guarantee wins; star power alone explained only a moderate amount of team success (r = 0.40).
- Overall team efficiency and shooting quality consistently mattered more than any single individual statistic.





Findings / Conclusion

- We were right that offensive efficiency and three-point accuracy are strong predictors of a winning team. Three-point percentage had a strong positive correlation with win percentage (r = 0.67), confirming that accurate shooting matters.
- We were partly wrong in expecting high-scoring players to mostly be on the top teams. The relationship between individual scoring and team success was only moderate (r = 0.40), meaning star power alone doesn't guarantee wins.
- The clearest takeaway is that winning teams succeed through efficient offense, strong overall team metrics, and accurate shooting—not just one dominant scorer.